Aston University

WEBIOS

Arab Academy
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BRITISH COUNCIL

EEG Signal Acquisition and Streaming

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HERBioLab

How does EEG measure changes in the brain?

Brain neurons are always active; however, EEG can't measure a neuron's small signal.

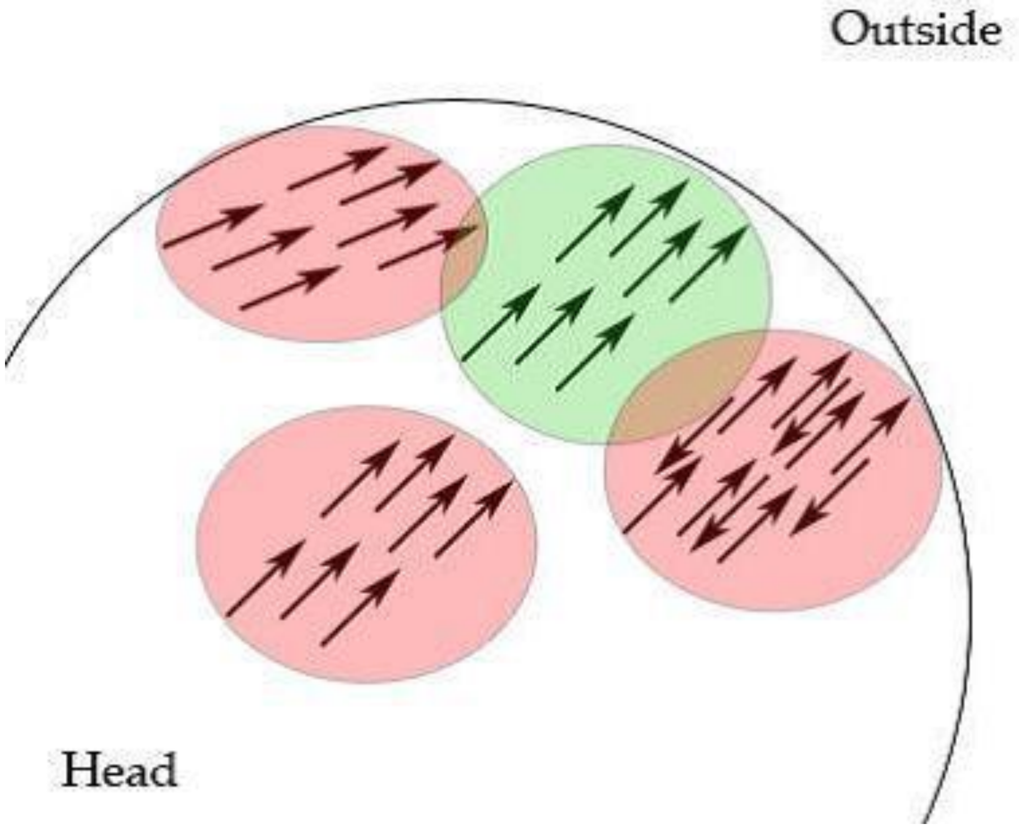
A large collection of neurons working together produces a large electrical current that can be detected by the EEG.

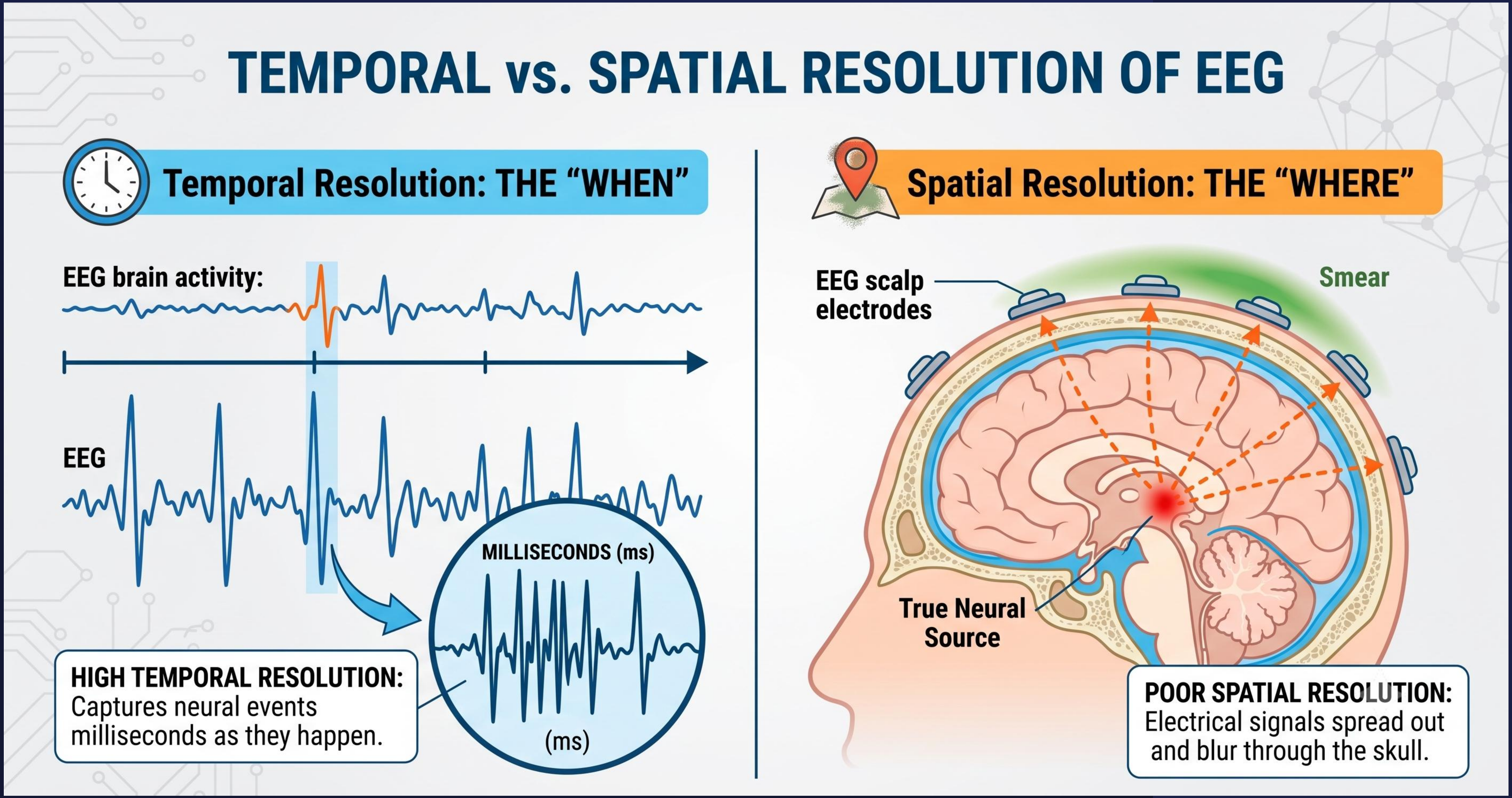
This requires certain conditions; hence, the EEG can't measure most neural activities.

How does EEG measure changes in the brain?

To achieve the best measurement of the signal by the EEG.

The group of neurons must produce an electric current perpendicular to the scalp.	The group of neurons must fire in parallel.	The group of neurons must fire with the same polarity, otherwise, they cancel each other out.

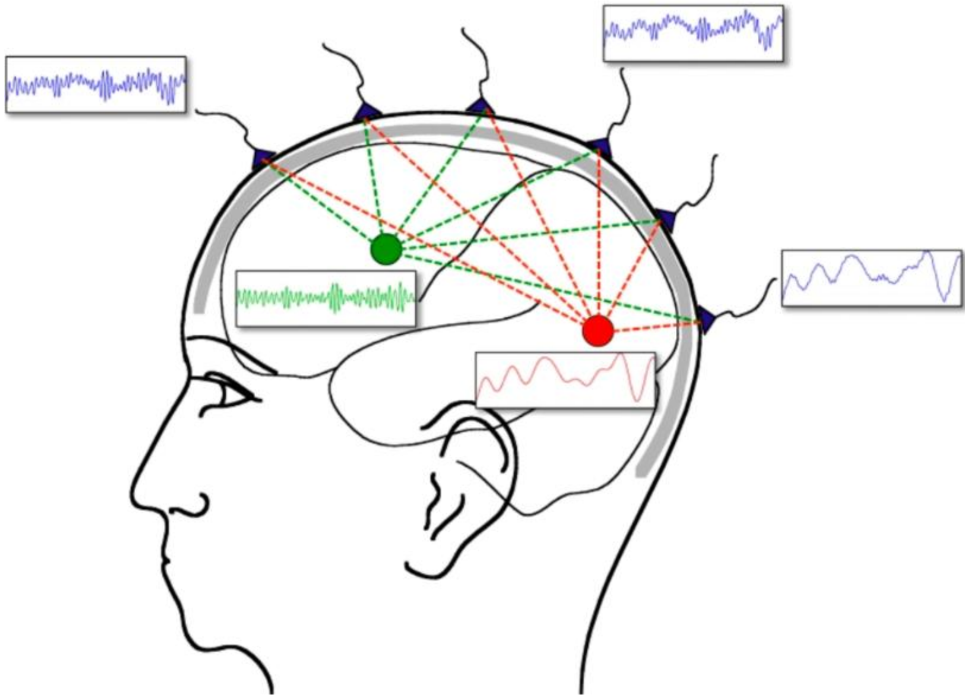



TEMPORAL vs. SPATIAL RESOLUTION OF EEG
Temporal Resolution: THE “WHEN”
EEG brain activity:
EEG
MILLISECONDS (ms)
(ms)
HIGH TEMPORAL RESOLUTION:
Captures neural events milliseconds as they happen.
Spatial Resolution: THE “WHERE”
EEG scalp electrodes
Smear
True Neural Source
POOR SPATIAL RESOLUTION:
Electrical signals spread out and blur through the skull.

Sensor-level versus source-level EEG analysis

Sensor-level
(Scalp
electrodes)

Source-level
(Reconstructed
underlying sources)



EEG recording

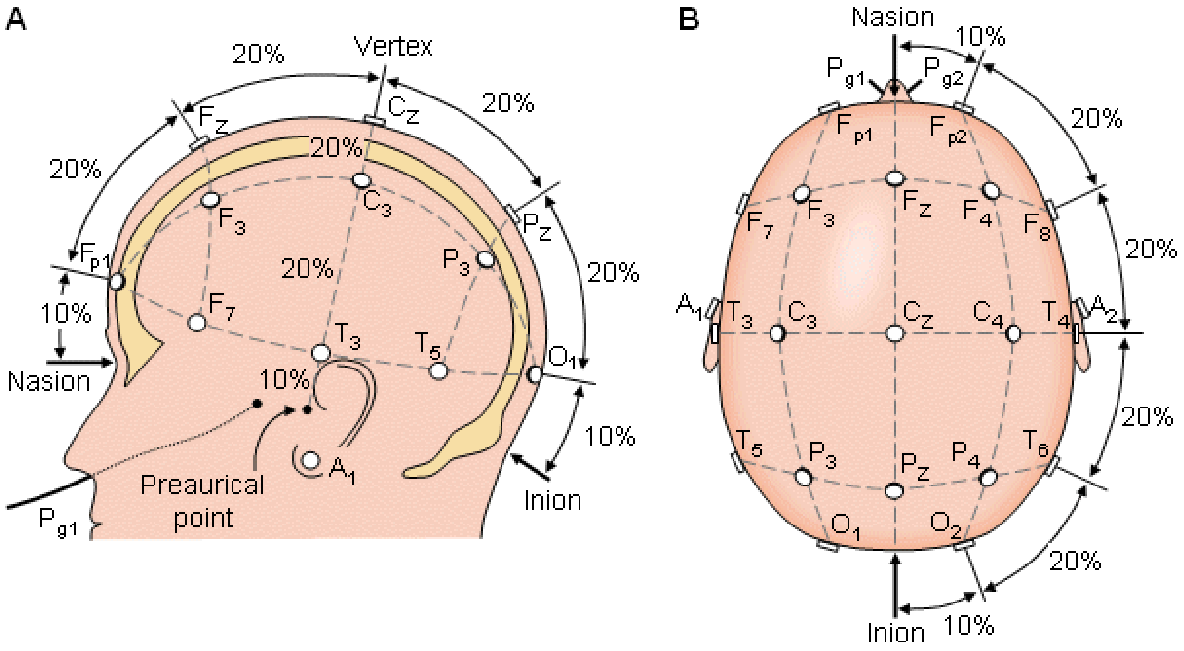
EEG uses high-conductivity electrodes (low impedance).

The 10-20 method separates electrodes by 10-20% of the head's circumference.

- Other systems have 32, 64, 128, 256, and 512 electrodes.

Previously, electrodes were put manually, which was time-consuming and impractical.

- Now, premade caps with the electrodes connected are used; these caps fit different patients.



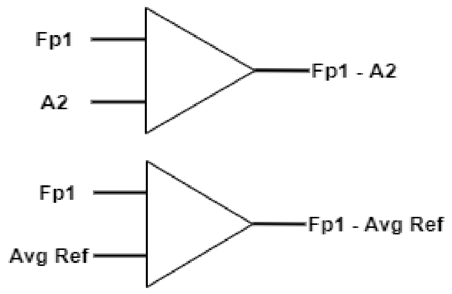
EEG montages

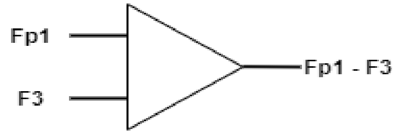
The recorded EEG signal is the difference between an electrode and a reference.

This set-up can be in two montages:

Monopolar

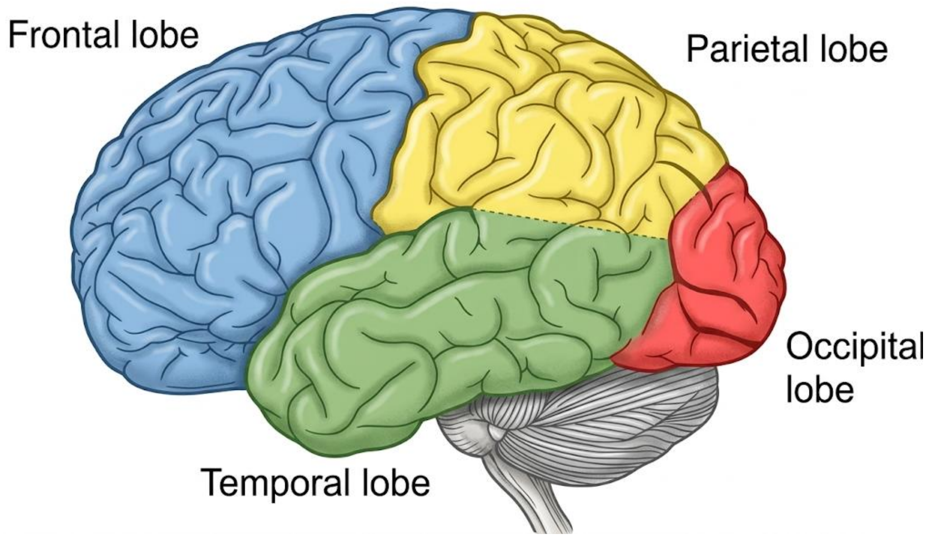
Bipolar





Odd numbers represent the left side of electrodes in standard notation, while midline electrodes are labelled by Z.

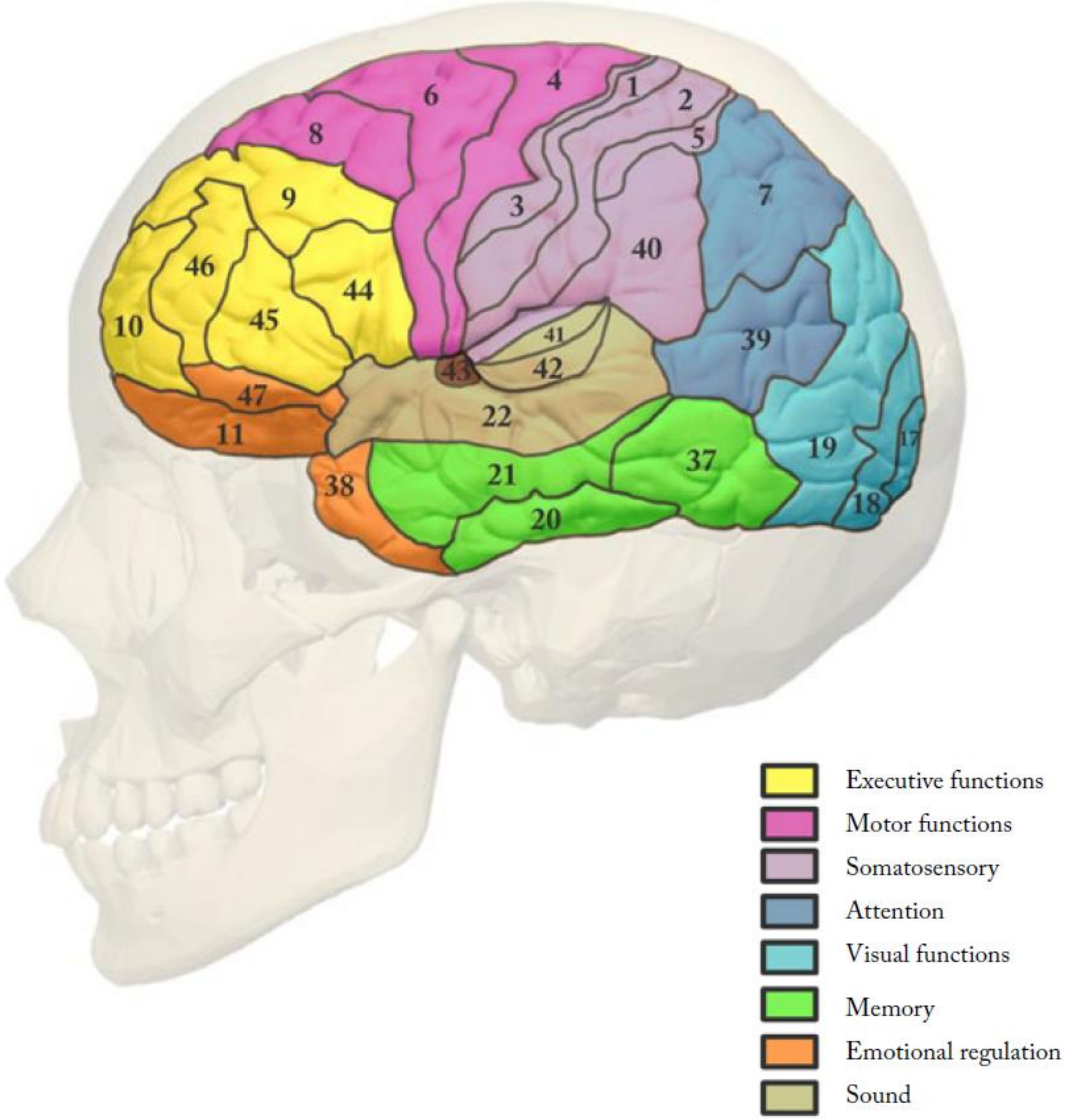
The letters F, P, T, O, and C signify electrodes above the frontal (F), parietal (P), temporal (T), occipital (O), and central (C) lobes, respectively.



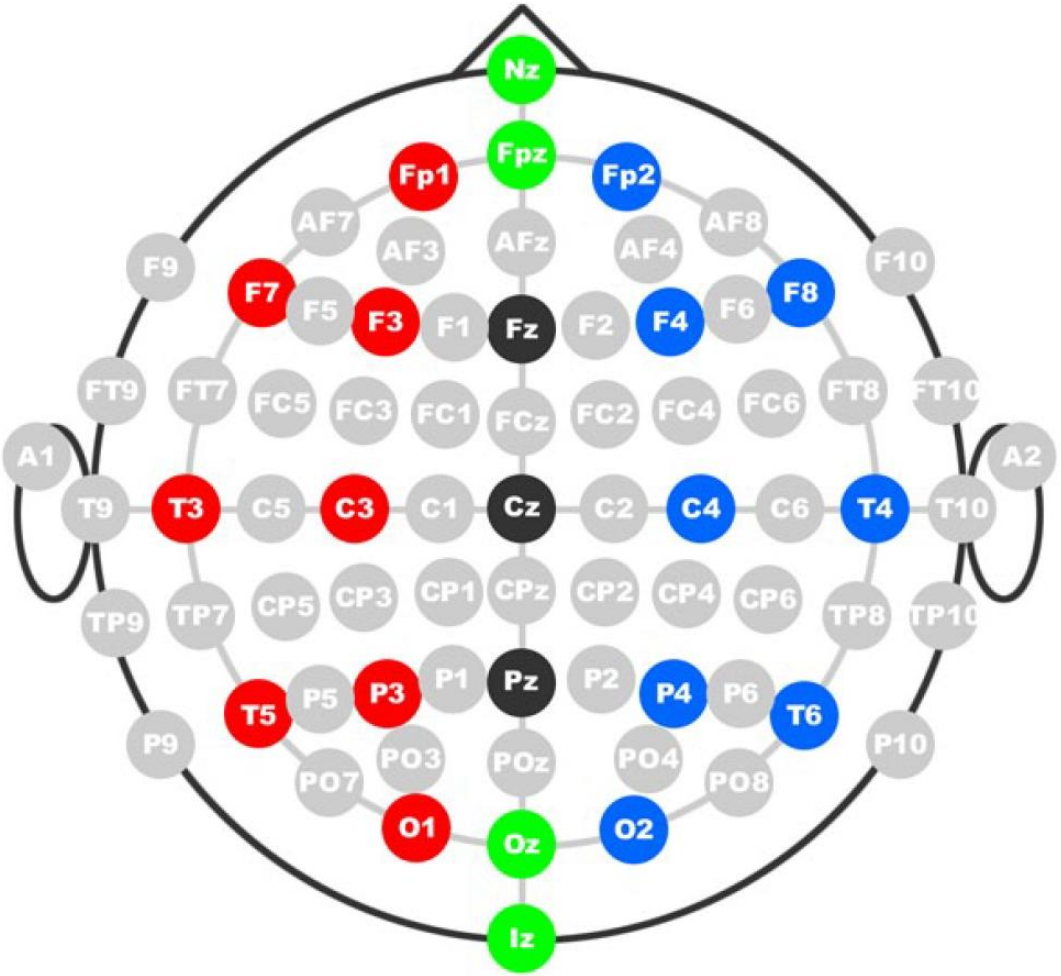
EEG measurement setups: Monopolar vs. Bipolar

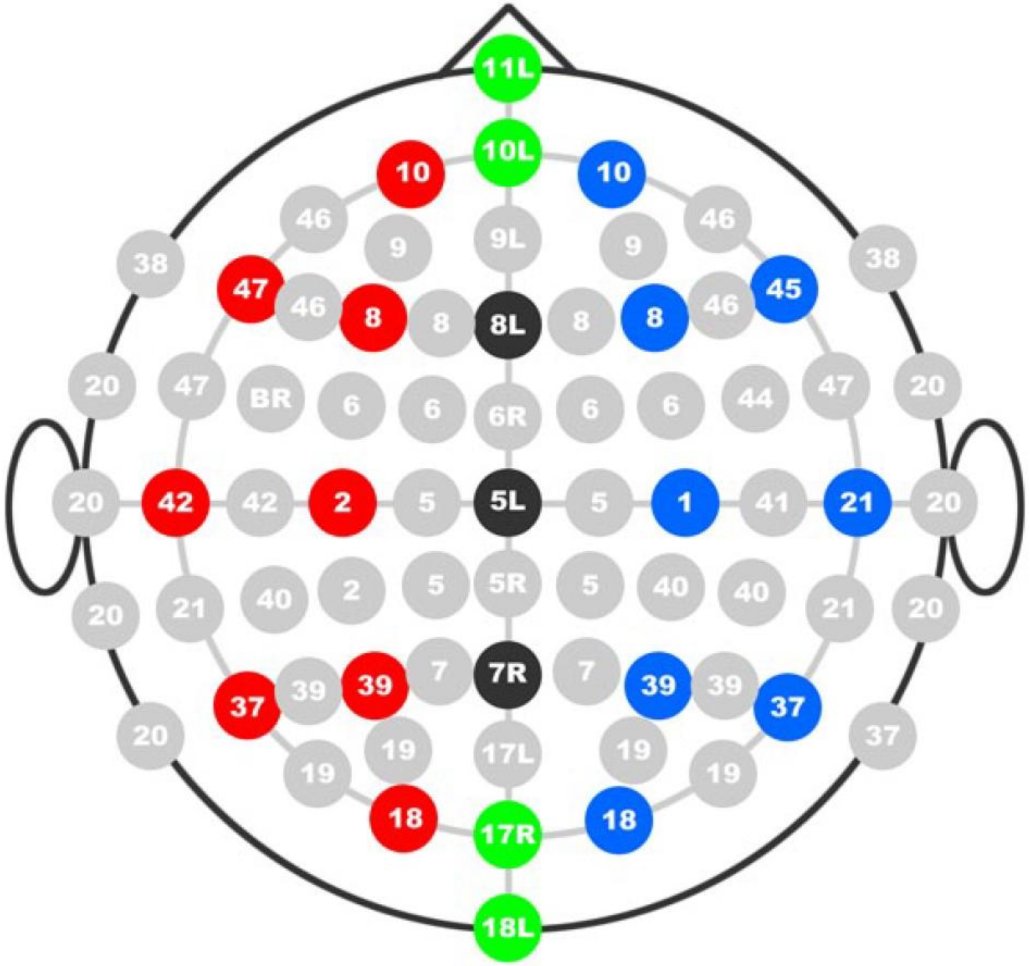
Feature	Monopolar (Referential) setup	Bipolar setup
Diagram layout	Electrode (**A**) on scalp connected against a reference electrode (**R**) on a neutral point.	Two distinct electrodes (**E1** and **E2**) placed directly on the scalp.
Amplifier logic	Voltage difference (**V**) = **A** - **R**	Voltage difference (**V**) = **E1** - **E2**
Measurement target	Measures voltage between the scalp electrode and a theoretically neutral body reference.	Measures the voltage difference directly between two scalp electrodes.
Best use case	Standard choice for mapping absolute brainwave amplitude across the head.	Highly effective for detecting localized focal activity.
Signal advantage	Isolates the true amplitude of a single scalp location.	Automatically cancels out shared, widespread background noise.

Brodmann areas



Electrode positions and corresponding Brodmann areas





EEG electrodes

Usage

- Electrodes sense scalp electric potential for EEG amplification and filtration circuits.

Precautions

- Good electrode-to-head contact is necessary for accurate EEG recordings.
- Electrodes are usually metal. The electrode-scalp impedance affects conductivity; thus, gel or paste is inserted between the electrode and the head. Some electrodes are gel-free (dry).

Fabrication

- Electrode caps with disposable electrodes (no gel or pre-gelled) are used in medical applications for hygienic precautions.
- Electrode caps with reusable disc electrodes (gold, silver, stainless steel, or tin) are used in research.

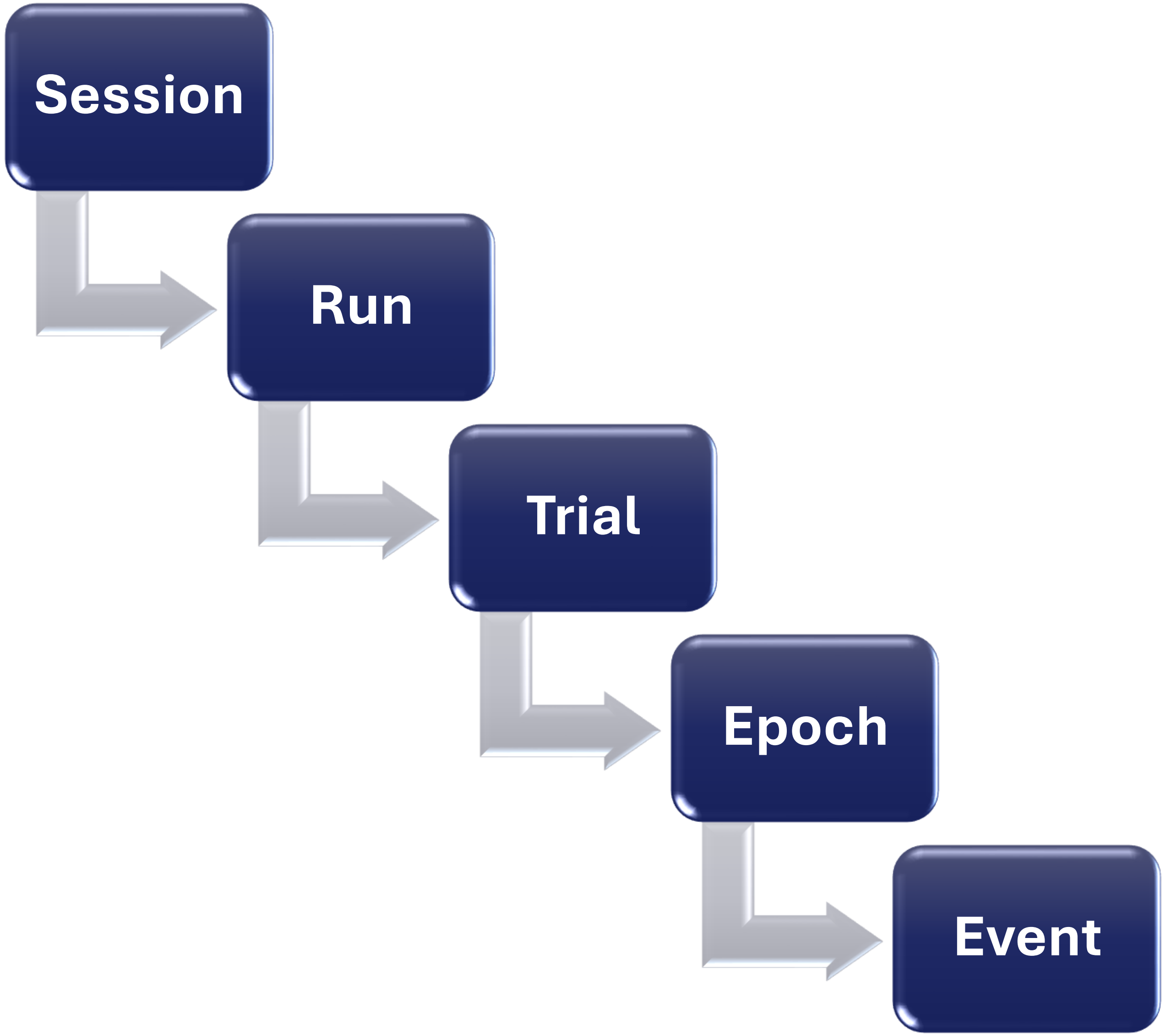
Active vs. passive EEG electrodes

Feature	Passive electrodes	Active electrodes
Hardware design	Standard simple disc (e.g., Ag/AgCl).	Features a built-in pre-amplifier attached directly to the scalp sensor.
Skin preparation	Requires physical scalp abrasion and conductive gel to work effectively.	Immediate signal boosting reduces the need for aggressive skin preparation.
Signal journey	The weak raw signal travels entirely through wires before reaching a distant main EEG amplifier.	The signal is immediately boosted at the source, sending a strong, robust signal down the wire to the main amplifier.
Vulnerability to noise	Highly susceptible to environmental interference (EMF) and wire movement, picking up significant noise.	The boosted signal effectively rejects environmental noise and movement artifacts during transmission.
Final data quality	Often results in noisy visual artifacts and poorer overall EEG data quality.	Delivers a highly clean, low-noise final EEG output.

EEG brain rhythms

Wave	Frequency (Hz)	Correlates with - Mental State	Correlates with - Location
Delta (δ)	2–4 Hz	Deep sleep	Center cerebrum and parietal lobes
Theta (θ)	4–8 Hz	Consciousness but drowsy	Positions not related to activity
Alpha (α)	8–14 Hz	Relaxation without paying attention	Occipital and parietal lobes
Mu (μ)	8–14 Hz	Voluntary movement	Motor cortex
Beta (β)	14–30 Hz	External active attention	Parietal and frontal lobes
Gamma (γ)	≥ 30 Hz	Associated with cognitive and memory	Somatosensory cortex

EEG-based experiment terminology



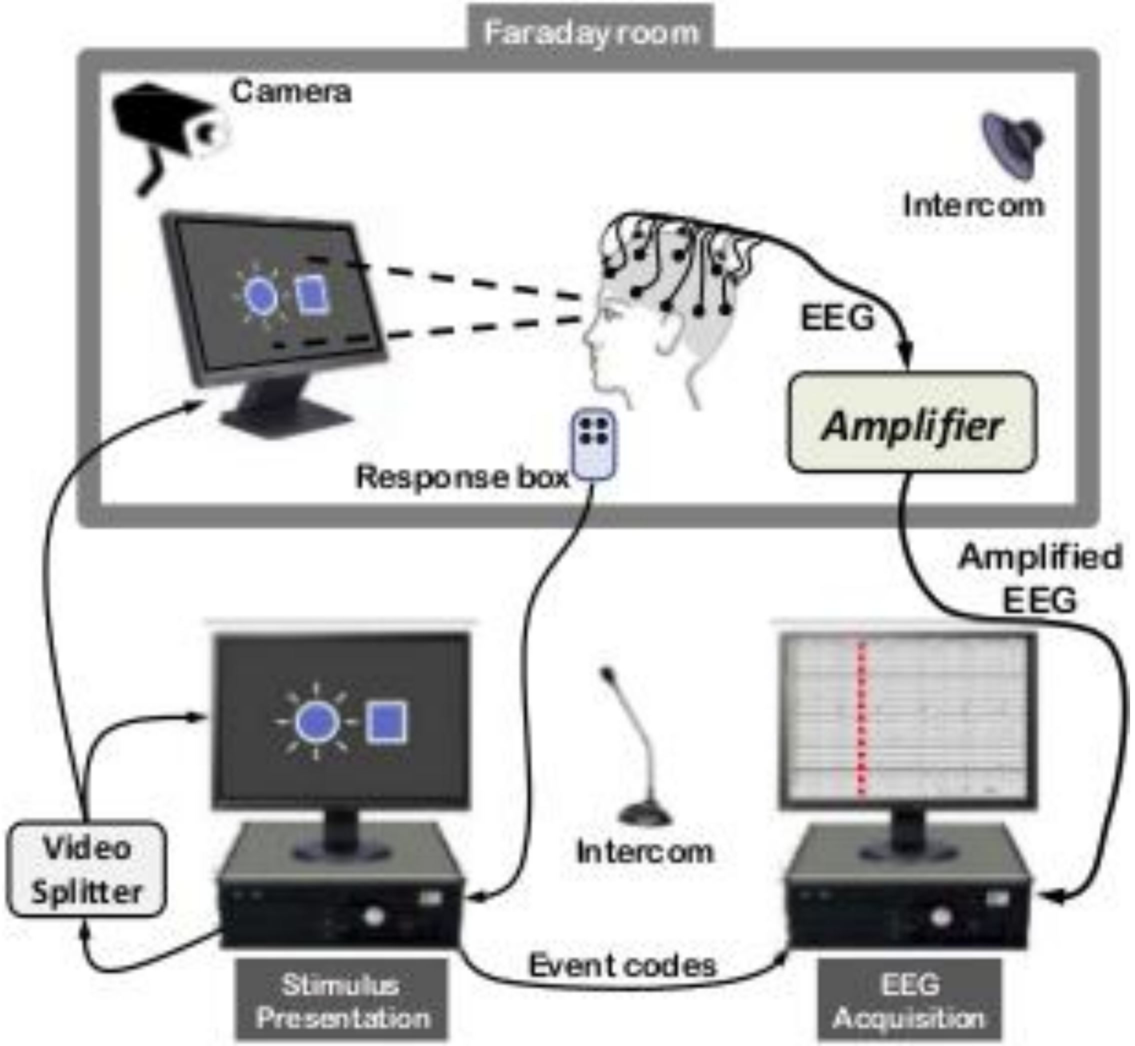
Experiment – Eligibility criteria of participants

General criteria

- Age
- Health
- Cognitive abilities
- Experimental conditions

Experiment – Recording setup physical components

- **Stimulus presentation**
- **Data acquisition system**
- **Computer system**
- **Shielding**
- **Air circulation**
- **Controlled lighting**



Experiment – Before official recording

Pre-tests

Pre-tests are preliminary testing of sequences, equipment and stimuli scripts performed in the EEG lab before data collection begins.

Pilot tests

EEG data from 2–5 people are routinely collected and evaluated to reveal errors, inferior design elements, or the modification of conditions.

Non-Invasive BCIs manufacturers

COMPUMEDICS NeuroScan

BRAIN PRODUCTS
Solutions for neurophysiological research

muse™
the brain sensing headband

NeuroSky
Body and Mind. Quantified.

EMOTIV

OpenBCI

g.tec

ant neuro
inspiring technology

BIOSEMI

Invasive BCIs manufacturers

Blackrock Neurotech

CLEARPOINT NEURO

NEURALINK

BRAINGATE TURNING THOUGHT INTO ACTION

synchron

neurable

OpenBCI

OpenBCI

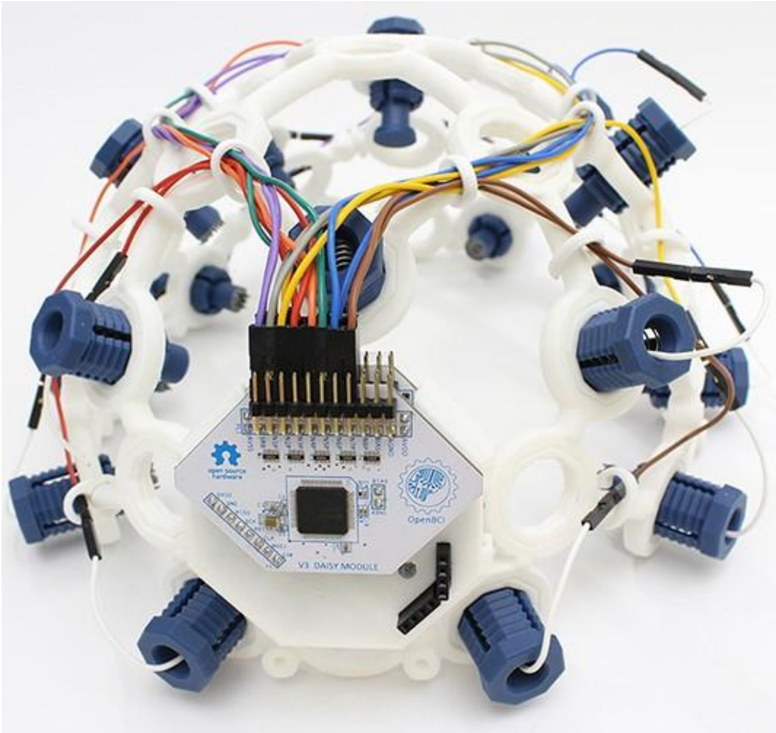
- **Website**: https://openbci.com/

Available biosensing approaches

- **EEG (Brainwaves)**: Captures cortical signals for paradigms like Motor Imagery, P300, and SSVEP.
- **EMG (Muscles)**: Tracks facial and peripheral muscle movements, such as jaw clenches and blinks.
- **ECG (Heart)**: Monitors cardiac rhythms, heart rate variability, and operator stress states.

Product highlights

- Unrestricted access to modify hardware architecture and software processing pipelines.
- Structural design files are free, allowing for localized manufacturing and custom anatomical fitting.
- Uses specialized comb sensors that navigate through hair, completely eliminating the need for conductive gels or skin abrasion.
- Modular setup supports 4-channel (Ganglion), 8-channel (Cyton), or 16-channel (Cyton + Daisy) configurations.
- Transmits via Bluetooth or Wi-Fi, granting freedom of movement while minimizing electrical line noise artifacts.



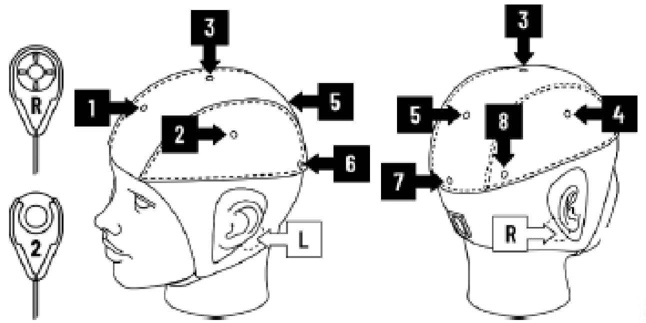
Unicorn Hybrid Black

g.tec Unicorn hybrid black

- **Website**: https://www.gtec.at/

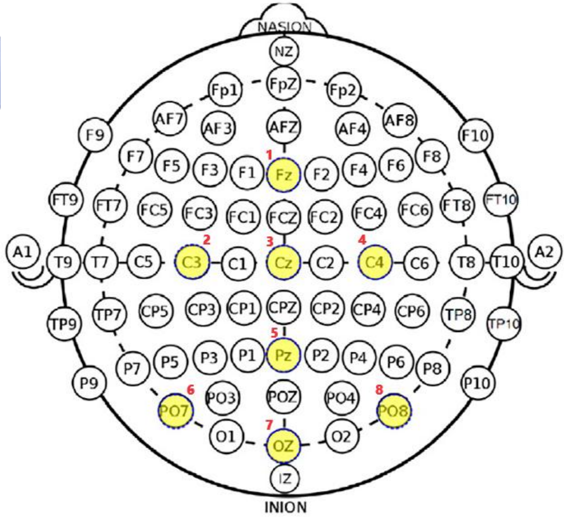
BCI approaches available within this headset

- **Motor imagery (MI) based BCIs**: Electrodes placed over the sensorimotor cortex.
- **P300 paradigms**: Electrodes positioned over the central, parietal, and occipital areas.
- **SSVEP and code-based VEP paradigms**: Electrodes located over the parietal regions.



Product highlights

- EEG recordings without cable connection via radio signal.
- Hybrid electrodes for wet and dry measurements.
- Correct positions of EEG electrodes for real brain wave recordings.
- 8 DC-coupled analog input channels with 24 Bit resolution.
- Sampling rate of 250 Hz per channel.
- Reference and ground EEG electrodes are securely fixed on the mastoids of the user, ensuring stable and accurate readings.



Muse 2

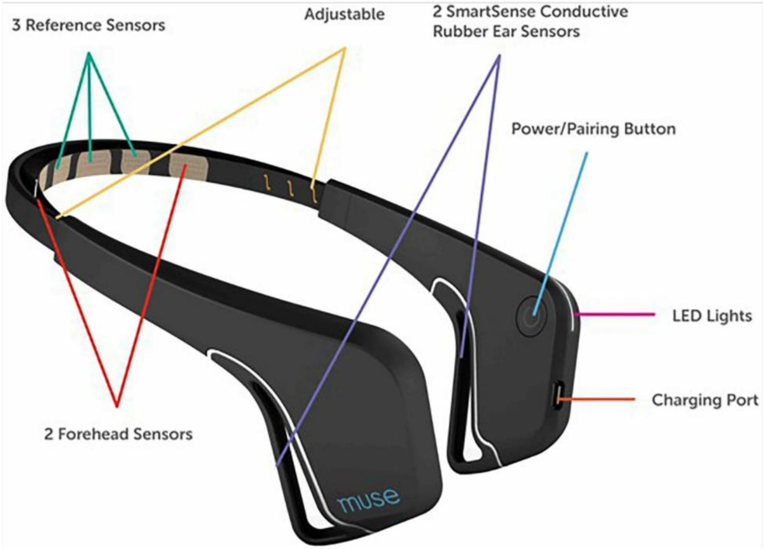
muse™
the brain sensing headband

Muse 2 brain-sensing headband

- **Website:** https://choosemuse.com/

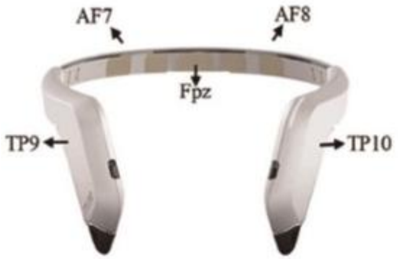
BCI approaches available within this headset

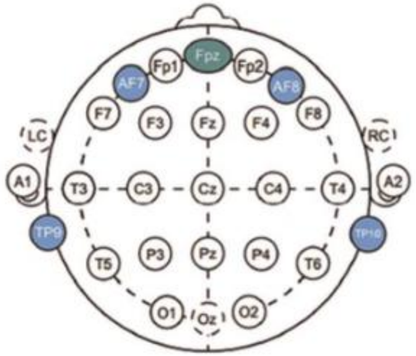
- **Active BCI approaches (Mental commands and intentional actions)**: A computer is trained to recognize specific, conscious mental states or actions.
- **Passive BCI approaches (Cognitive state monitoring)**: Instead of using your brain to control an interface, the interface adapts based on your cognitive state.



Product highlights

- 4 EEG channels (detect brainwaves) + 2 auxiliary channels
- PPG (heart-rate), breathing and movement sensors (accelerometer)
- EEG sample rate: 256 Hz, resolution 12-bit
- Electrode positions: TP9, AF7, AF8, TP10; reference electrode FPz
- Accelerometer: three-axis @ ~52Hz, 16-bit resolution





Open-source software for task design and processing

While setting up an experiment one must decide which software to use for stimulus presentation.

- The following table gives a short overview of the programs that can aid in designing your task.
- If none of them is applicable to what you plan to do, then you need to write a **script**.

sccn/BCILAB

MATLAB Toolbox for Brain-Computer Interface Research

	Primary Language	Python Bindings	BCI focused
BciPy	Python	yes	yes
BCI2000	C++	yes	yes
OpenViBE	C++	yes	yes
PsychoPy	Python	yes	no
Psychtoolbox	Matlab	yes	no
PyFF	Python	yes	no
BCILAB	Matlab	yes	yes

Muse 2 headset connection

Pre-flight hardware checks

Connection with muse 2 using mobile Apps

muse

The Official Muse App: Consumer UX

- Converts raw EEG into immediate, gamified audio biofeedback (e.g., weather sounds).
- Hides complex frequencies behind simple consumer metrics like "calm" vs. "active" states.
- Combines EEG with PPG and accelerometer data for holistic breath and posture tracking.
- Showcases a closed-box UX designed for public accessibility rather than BCI development.

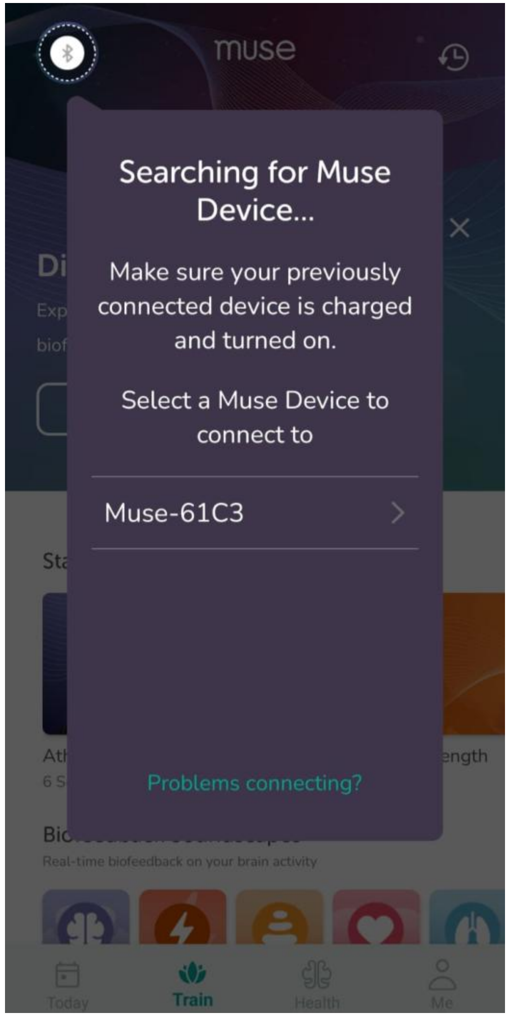
Mind Monitor: The Developer Toolkit

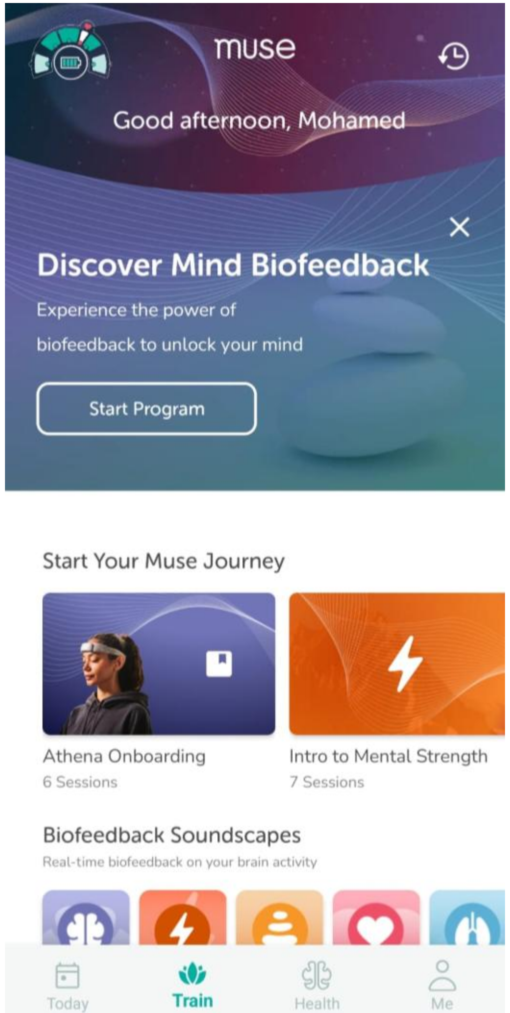
- Exposes the raw microvolt readings and real-time spectrograms that commercial apps hide.
- Visually isolates specific brainwaves (Delta, Theta, Alpha, Beta, Gamma) for cognitive state tracking.
- Allows direct recording of synchronized EEG and motion data to CSV for offline analysis.
- Streams live data wirelessly via OSC, turning the phone into a bridge for PC software.

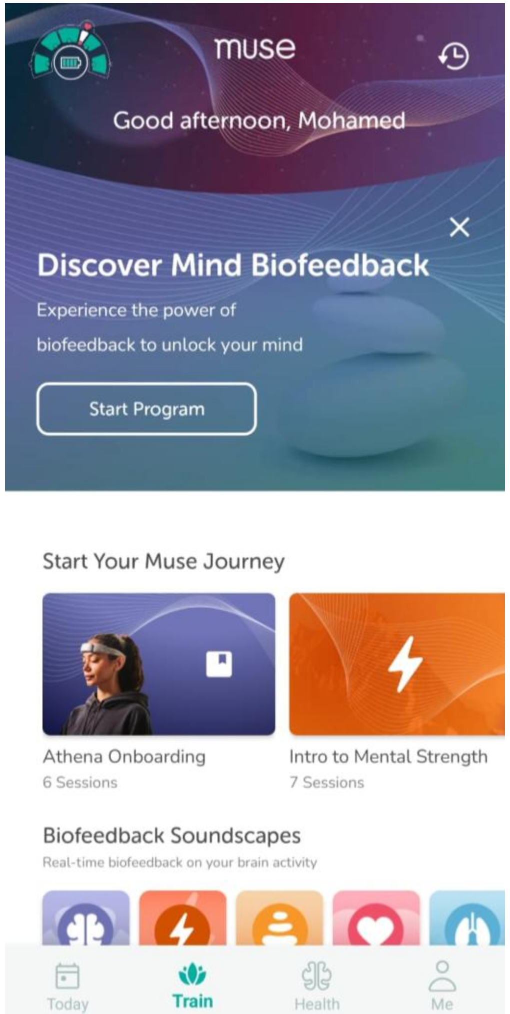
Myndlift: Clinical Application

- Upgrades off-the-shelf consumer hardware into a clinical-grade therapeutic tool.
- Facilitates customized neurofeedback training monitored remotely by healthcare professionals.
- Provides a working model of a closed-loop system used for continuous cognitive training.
- Illustrates the jump from consumer neurotechnology to meeting rigorous medical sector standards.

Connecting using the Muse official app







The neural engineer's mandate: Concepts over code

The mission
- Master neural dynamics, not software development.

The tool
- Code is simply a vehicle used to test the scientific hypotheses.

The strategy
- It is completely valid to leverage AI or software engineers to build the infrastructure and write the syntax.

The responsibility
- You must understand the underlying signal processing and ML to prove the system is scientifically sound.

System setup: PyCharm environment and dependencies

Workspace initialization

- **IDE configuration**: PyCharm Professional or Community Edition.
- **Python interpreter**: *Python 3.9 to 3.11* is recommended to ensure stable compatibility with the mne and pylsl packages.
- **Data root directory**: The pipeline defaults to writing to *"Directory"*. You must physically create this folder on your machine or update the BASE_DATA_DIR variable across all scripts to match your local path.

Core dependency installation

- Open the PyCharm integrated terminal and install the specific BCI software stack:
  - pip install numpy pandas scipy scikit-learn (Math and machine learning)
  - pip install mne (Signal processing and ICA artifact scrubbing)
  - pip install pylsl muselsl (Hardware streaming via Lab Streaming Layer (LSL))
  - pip install pygame joblib (Visual UI engine and model serialization)

Hardware prerequisites

- **Native Bluetooth low energy (BLE)** enabled on the host Windows machine.
- **Muse 2 headset fully charged**, turned on, and actively in pairing mode before executing the scripts.

Connecting using the script

os	Allows the script to interact with your operating system to verify it is running on Windows and to locate its own unique process ID.
threading	Enables the script to run a simultaneous background timer that pulls the viewer window to the front without freezing the main program.
asyncio	Secures an asynchronous event loop, which is often required under the hood for stable Bluetooth Low Energy (BLE) connections.
sys and time	Used for executing system-level exits, managing operational delays, and executing cursor-level console output overwrites for the live dashboard.
multiprocessing	Allows the script to isolate the Bluetooth connection into a background daemon process so the main terminal remains responsive.
numpy (np)	Utilized for rapid mathematical array operations, specifically calculating the peak-to-peak amplitude of the brainwaves.
pylsl	Handles the Lab Streaming Layer (LSL) protocol to locate and ingest the live EEG data from the local network.
muselsl	The **core** library utilized to bridge the connection to the Muse headset, broadcast the data, and launch the real-time visualizer.

Automated pre-flight pipeline overview

To ensure data integrity before recording, the script executes a structured, five-step hardware verification pipeline.

Background stream initialization

- A direct Bluetooth link is established with the headset, and data is broadcasted to the local machine.

Network resolution and handshake

- The system actively scans the local network to locate the newly created EEG broadcast and binds to it.

Continuous signal diagnostics

- A looping dashboard evaluates the real-time electrical contact quality of all four electrodes against calibrated thresholds.

Real-time visualization launch

- Regardless of whether the diagnostic passes or times out, a visual oscilloscope is launched for manual inspection.

Safe termination

- The background Bluetooth stream is safely severed to prevent adapter crashes once the operator closes the viewer.

Targeting the headset by MAC in two ways

A hard-coded **MAC address** lets the script reconnect to the same headset directly, skipping the slower discovery scan.

Leave it empty “**None**” and the program instead scans, prints every Muse it finds, and exits so you can copy an address.

Bluetooth connection safety net (asyncio)

Python needs an active "event loop" to handle Bluetooth, but background tasks often start without one.

The script checks if an event loop is already running (get_event_loop).

If it cannot find a loop, it catches the resulting error to prevent the program from crashing.

It automatically builds a brand-new loop (new_event_loop) and attaches it, guaranteeing the headset connects successfully.

Hardware initialization (start_stream)

Skips the slow Bluetooth search by using the headset's exact ID (TARGET_MAC_ADDRESS) to connect instantly via stream().

If the specific ID is missing, the system switches to discovery mode (list_muses()) to scan and list all nearby headsets.

Safely stops the script (sys.exit()) if no devices are found, preventing the program from freezing or hanging indefinitely.

The core diagnostic engine (run_preflight_check)

Locates and hooks into the live EEG stream over the local network (resolve_byprop, StreamInlet).

↓

Clears out old, stale Bluetooth data packets (pull_chunk) to ensure only real-time data is evaluated.

↓

Slices the signal into 1-second chunks and calculates the peak-to-peak voltage (np.ptp) across every channel.

↓

Compares amplitudes against safety limits, flagging channels as FLATLINE (< 2.0 μV) or NOISY (> 350.0 μV).

↓

Continuously refreshes the terminal screen to show an auto-updating status table.

Orchestration and visualization (__main__ and start_viewer)

Runs the Bluetooth connection in an isolated background process (multiprocessing), ensuring it terminates cleanly if the main program crashes.

↓

Launches the stream, waits 3 seconds for the network stream (LSL) to stabilize, and triggers pre-flight checks.

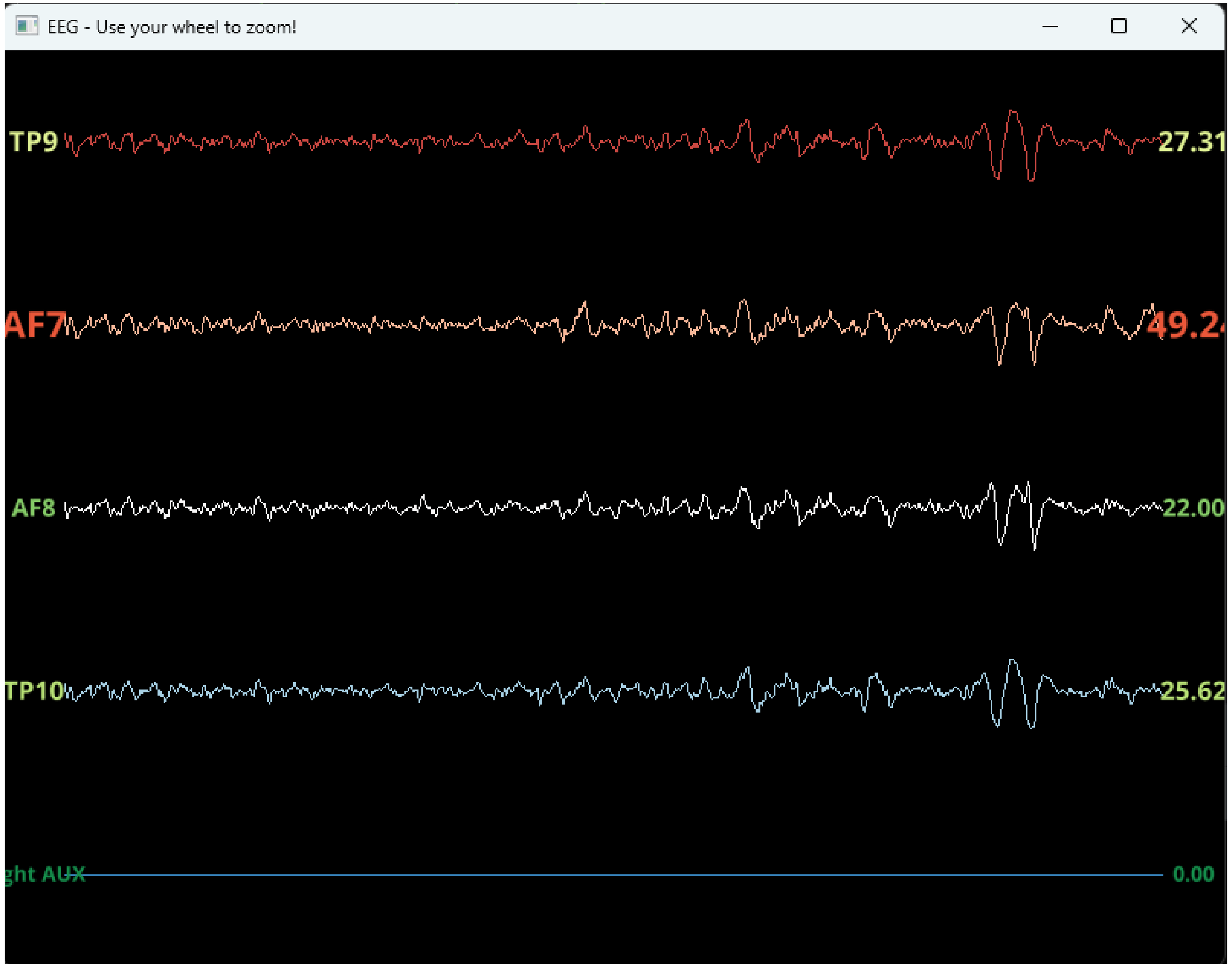
↓

Opens the real-time EEG visualizer window and brings it directly to the front screen.

↓

Closing the viewer safely shuts down the stream process (stream_process.terminate()), preventing Bluetooth adapter freezes.

Muse live streaming



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Thank You!

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